The Mariner's Mirror

The International Quarterly Journal of The Society for Nautical Research

ISSN: 0025-3359 (Print) 2049-680X (Online) Journal homepage: http://www.tandfonline.com/loi/rmir20

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To cite this article: Simon C. Davidson (2016) The Use of Chronometers to Determine Longitude on East India Company Voyages, The Mariner's Mirror, 102:3, 344-348, DOI: 10.1080/00253359.2016.1210326

To link to this article: http://dx.doi.org/10.1080/00253359.2016.1210326

Published online: 19 Jul 2016.

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References

Arbex, J. C., 'La Vela al Tercio y los Últimos Pesqueros a Vela del Golfo de Vizcaya', *Itsas memoria. Revista de Estudios Marítimos del País Vasco*, 2, (1998), 363–78

Blanco-García, J., *Guía de las Embarcaciones Tradicionales Gallegas* (Vigo, 2009)

Cadoret, B., D. Duviard, J. Guillet and H. Kérisit, *Ar Vag* (Douarnenez, 2003)

Ferreira, E., *Galicia en el Comercio Marítimo Medieval* (Corunna, 1988)

Garret, R., *The Symmetry of Sailing* (New York, 1996)

Gilles, D., and M. Thersiquel, *Gréements des Côtes de France* (Paris, 1996)

McGrail, S., *Ancient Boats in N.W. Europe* (New York, 1987)

Mörling, S., *As Embarcacions Tradicionais de Galicia* (Santiago de Compostela, 1989)

—, *Lanchas y Dornas* (Santiago de Compostela, 2005)

Oller, F. and V. García-Delgado, *Nuestra Vela Latina* (Barcelona, 1996)

Palmer, C., 'Windward Capabilities of Ancient Vessels', *The International Journal of Nautical Archaeology*, 38.2 (2009), 314–30

Sleeswyk, A. 'The Ever or Boene Rinses and the origin of the Lug sail in N. W. Europe', *Mariner's Mirror*, 72:1 (1986), 78–81

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The Use of Chronometers to Determine Longitude on East India Company Voyages

John Arnold in his *Certificates and Circumstances* states that various East India Company captains used his chronometers. The earliest instance quoted is that of Captain Charles Mears on board HEICS *Egmont*. Captain Mears wrote to him from the Downs off Margate, concerning the use of one of his timekeepers on 6 March 1773.[1] His letter records the longitude using the timekeeper while anchored off the Mother Bank at the Downs with seven readings over the period 3 to 5 March. The letter meticulously records in a tabular form the correction he applies to the rate of going which was provided by John Arnold of +13 seconds gaining per day. From his tabular results enclosed in the letter, it is also clear that his use of the word 'timekeeper' means a box chronometer, as he was using a watch to transfer the time of observation to the timekeeper. This has, until now, been regarded as the first use of a chronometer by a captain on a regular commercial voyage. There is, however, evidence that Mears used a chronometer to determine longitude on an earlier voyage in 1770.

In his log for the 1769/70 voyage aboard the *Egmont*,[2] he made the following annotation in his own handwriting on the flyleaf:

> 13 March 1770 at 21h 30' 38" Meantime, the Timekeeper was then 15" slower than meantime in the Roads of St. Cruz, Isle of Teneriffe.
>
> March 16 at 18h 44' 22" meantime the Timekeeper was 1' 12" slower than meantime same place.[3]

Mears was checking the mean time recorded by his timekeeper against the known longitude and therefore derived the mean time of Santa Cruz while at anchor in the Roads off Santa Cruz, Tenerife. Unfortunately, he does not mention the name of the maker; however, we can assume with a high degree of certainty that at this date it would have been made by John Arnold. It is possible that it might have been

1 Arnold, *Certificates and Circumstances*, 3; Mercer, John Arnold, 103–5.

2 A voyage identified as for example 1772/3 means the ship was signed up and given its sailing orders in 1772, was dispatched some time after November to reach its destination in 1773 and would arrive back towards the end of 1774.

3 British Library (BL), India Office Records (IOR), L/MAR/B/535F, Egmont, Captain Charles Mears's Log 1769/70. The Captain's Logs of East India Company chartered ships do not show any record of the use of a chronometer in any of the voyage seasons from 1762/3 to 1768/9.

Arnold's first attempt at a marine timekeeper, the detached cylinder timekeeper of about 1768.[4]

The other 29 ship's logs of the 1769/70 voyage season make no mention of the use of a timekeeper and it is possible that Mears acquired his through a personal connection with Arnold. In all the certificates quoted by Arnold in his *Certificates and Circumstances*, it is only that of Mears that contains a personal message about him and his family. Mears's second letter indicates that he and Arnold were friends and it was not just a business relationship. At the end of Mears's letter concerning the steady rate of his Arnold watch and the loss of his rating records on the wrecking of his ship HEICS *Brilliant* he adds:

> I hope you and your family enjoy perfect health, and that you will soon acquire a fortune, by serving, at the same time, all Navigators of Great Britain and indeed I think the nation ought to reward you: for, notwithstanding the certainty and facility of our finding the longitude by the Lunar Observations, yet there happens many times when we can know our longitude by the Time-keeper, though the Moon is not then to be seen.[5]

Mears's record is a year before Arnold delivered a timekeeper to Rear-Admiral Sir Robert Harland aboard HMS *Northumberland* at Spithead on 4 March 1771 which the admiral records using in July 1771 when sailing between the Cape of Good Hope and Madagascar.[6] It also predates by two years the second voyage in July 1772 of Captain Cook when he was equipped with various timekeepers by the Board of Longitude.

The evidence from the 86 ship's logs for the three succeeding voyages following 1769/70 indicates that, apart from Mears, the only other captain to use a timekeeper was Captain Nicholas Skottowe on HEICS *Bridgewater*. He recorded in his log on 11 March 1773: 'The Timepiece was sett 28 January and said to lose 1½" a day uniformly is always to be allowed for in the Observations.'[7] Later, on 10 April, when recording the longitude by the timepiece he stated:

> At 7h 46' 32"AM x time 1h 4' 43" = 16h 10' 3" West of Greenwich which I shall keep in future, the timepiece being regulated to Greenwich time by a Note I have from the Maker.

Early use of the lunar-distance method on a commercial voyage

We will see below that Captain Mears was accomplished in 'finding the longitude by the Lunar Observations'. This was the lunar-distance method (lunars), which used the moon's apparent motion relative to the sun or stars like a clock to find a reference time, e.g. the time at Greenwich. The observer used an octant, later the sextant as it became more widely available, to measure the angle between the moon and star or sun and their heights. The method involved considerable calculation following the observation to make allowance for atmospheric refraction, owing to bending of light rays in the atmosphere and for lunar parallax, because the observations are made from the earth's surface not its centre whereas the tables of the position of the moon refer to the centre of the earth. The observer then consulted prepared tables of lunar distances and the times at which they will occur. The difference between this time and local time on board the ship which is determined from the altitude of the sun or star and can be converted into a difference of longitude, one hour of time difference equating to 15° of longitude. Nevil Maskelyne and his assistant, Robert Waddington, had both tried out tables developed by Tobias Meyer on a voyage they made in 1755 to and from St Helena and separately published tables of lunar distances on their return in 1763.[8]

4 Gould, *The Marine Chronometer*, 2nd edn, 107, n1; Betts, 'Arnold and Earnshaw', 314.

5 Arnold, *Certificates and Circumstances,* 39.

6 Gould, *The Marine Chronometer*, 2nd edn, 106, see n**; Harland had two Arnold timekeepers on board, one owned by the Board of Longitude and one owned by the admiral. Mercer, John Arnold, 22–4.

7 BL, IOR, L/MAR/B/42F Bridgewater, Captain Nicholas Skottowe's Log, 1772/3.

8 Maskelyne, *The British Mariners' Guide*; Waddington, *A Practical Method*.

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Derek Howse mentioned that four East India officers had testified in support of the lunar-distance method for determining longitude at the 9 February 1765 meeting of the Board of Longitude, which was considering a long memorial on this subject from Nevil Maskelyne.[9] In fact on inspecting the minutes of that meeting one finds that two East India Captains from HEICS *Egmont* and *Speaker* gave written testimony that they had successfully used the method published by Maskelyne in *The British Mariners Guide* on their last voyage just ended in the season 1762/3.[10] Their testimony was supported by their officers showing their workbooks to the Board at this meeting.[11] In addition John Horsley, Fourth Officer in the *Glatton*, also showed his workbook but there was no testimony from his captain and the captain's log showed no sign of longitude determination by lunars so clearly this officer was acting on his own.[12]

What is significant about this is that Mears was one of the two East India Company captains who gave written testimony on the successful use of the method. Clearly he was very interested in utilizing the new technique promulgated by Maskelyne and Robert Waddington. Waddington noted that he had demonstrated his method to 'several of the chief and second commanders of ships in the employ of the East India Company' and that he had had 'the pleasure of instructing some of those gentlemen in the method'.[13] It would be fair to assume that Mears was one of these gentlemen.

It is apparent from the Board of Longitude's minutes that the captains and officers provided their testimony at Nevil Maskelyne's request. Mears and Captain Dewar had both returned in 1764 from voyages to India of approximately 18 months' duration. Maskelyne must have heard that they had been using the lunar-distance method throughout their voyages. We know that Maskelyne was well connected with the East India Company through his sister who was the wife of Lord Robert Clive and it is not inconceivable that he attended the New Jerusalem Coffee House, a meeting point for East India Captains, to learn if any were using his lunar-distance method to determine longitude.

John Arnold, who was in contact with Maskelyne, would have heard which captains were using the lunar-distance method, which would indicate to him captains interested in new methods of determining longitude. It would also be useful to test the accuracy of any chronometer while at sea to determine if the chronometer was changing its rate. This is almost certainly how Mears either bought or was supplied with an Arnold chronometer in 1770 and 1773.

In the logs for the voyage seasons following that of 1762/3 both Mears and Dewar continued to use lunars and two of their officers, John Lauder and Robert Scott, who would become captains in due course, used the lunars in the ships they commanded. Apart from these four captains with prior experience only two other captains, John Mitford and William Smith, used lunars in the voyage seasons up to and including 1768/9 (see table 1).

In the ship's logs of the East India Company for the four subsequent voyage seasons of 1769/70, 1770/1, 1771/2 and 1772/3 it can be seen that respectively 30, 9, 22 and 52 per cent of ships used the lunar-distance method for calculating longitude. The lumpy progression of take-up is explained by the fact that those ships using lunars in 1769/70 would only return to England in late 1771 or the first half of 1772. This means the communication of their successful use of lunars to other captains would only benefit in reality those setting out in 1771/2 or more likely 1772/3.

It seems that Mears and Dewar together with their two officers, Lauder and Scott, had

9 Howse, 'The Lunar Observation Method', 155.

10 BL, IOR, L/MAR/B/535D *Egmont*, Capt. Charles Mears's Log, 1762/3 and L/MAR/B/548B *Speaker*, Capt. James Dewar's Log, 1762/3.

11 Cambridge University Library, Board of Longitude Fair Minutes RGO 14/5, 83–5. First Officer John Lauder of HEICS *Egmont* (1); Third Officer Robert Scott and Sixth Officer James Stephens of HEICS *Speaker.*

12 BL, IOR, L/MAR/B/172A *Glatton* Capt. Richard Doveton's log 1762/3.

13 Waddington, *A Practical Method*, ii.

Table 1 Analysis of longitude determination by voyage season

Voyage season	*Ship*	*Captain*	*Longitude Method*	*Ref: IOR L/MAR/B*	*No. ships per season*
1762/3	*Egmont* (1)	Charles Mears	Lunar	535D	21
"	*Speaker*	James Dewar	Lunar	548A	"
"	*Glatton*	Richard Doveton*	Lunar	172A	"
1765/6	*Speaker*	James Dewar	Lunar	548B	23
1766/7	*Egmont* (2)	Charles Mears†	Lunar	535E	25
"	*Houghton*	William Smith	Lunar	438K	"
"	*Northumberland*	John Mitford	Lunar	141D	"
1767/8	*Duke of Gloucester*	John Lauder‡	Lunar	547B	27
1768/9	*Speaker*	Robert Scott¶	Lunar	548C	33
1769/70	*Duke of Gloucester*	John Lauder	Lunar	547C	30
"	*Egmont* (2)	Charles Mears	Lunar/Timekeeper	535F	"
"	*Hawke*	John Cotton	Lunar	390E	"
"	*Houghton*	William Smith	Lunar	438L	"
"	*Latham*	John Prince	Lunar	482E	"
"	*Northumberland*	John Mitford	Lunar	141E	"
"	*Princess Royal*	Robert Kerr	Lunar	405D	"
"	*Valentine*	Charles Purvis	Lunar	452D	"
"	*Worcester*	Richard Hall	Lunar	278E	"
1770/1	*Deptford*	William Tryon	Lunar	426D	32
"	*Sea Horse*	Edward Dampier	Lunar	479B	"
"	*Queen*	George Staniforth	Lunar	356D	"
1771/2	*Duke of Grafton*	Samuel Bull	Lunar	508B	27
"	*Fox*	David Mitchell	Lunar	456F	"
"	*Hector*	Stephen Williams	Lunar	486H	"
"	*Royal Charlotte*	John Clements	Lunar	150D-E	"
"	*Royal Henry*	Robert Rous	Lunar	476A	"
"	*Speaker*	Robert Scott¶	Lunar	548D	"
1772/3	*Bridgewater*	Nicholas Skottowe	Lunar/Timekeeper	42F	27
"	*British King*	Daniel Hoare	Lunar	557D	"
"	*Duke of Kingston*	Raymond Snow	Lunar	545C	"
"	*Egmont* (2)	Charles Mears	Lunar/Timekeeper	535G	"
"	*Houghton*	William Smith	Lunar	438M	"
"	*Kent*	David Thompson	Lunar	317K	"
"	*Lord Mansfield*	William Fraser	Lunar	463E	"
"	*Prime*	James Dundas	Lunar	509B	"
"	*Princess Royal*	Robert Kerr	Lunar	405E	"
"	*Resolution*	Thomas Poynting	Lunar	466B	"
"	*Royal Captain*	Edward Barrow	Lunar	556E	"
"	*Valentine*	James Ogilvie	Lunar	452E	"
"	*Vansittart*	Robert Young	Lunar	46D	"
"	*Worcester*	John Cook	Lunar	278F	"

Notes to table overleaf

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* Fourth Officer John Horseley's workbook shown to Board of Longitude meeting of 9 February 1765, Captain Doveton not involved.
† Captain Mears missed a voyage season in 1765/6 as awaiting the launch of *Egmont (2)* to replace the condemned *Egmont (1).*
‡ John Lauder was First Officer on *Egmont (1)* 1762/3 voyage.
¶ Robert Scott was Third Officer on *Speaker's*1762/3 voyage and Second Officer on *Speaker's* 1765/6 voyage.

created great interest in the lunar-distance method by the start of voyage season 1769/70. It may not have been a coincidence that the Court of Directors of the East India Company resolved on 7 October 1768 that in future all officers and mates would be required to produce certificates vouching that they were qualified in the method of lunar observation before being examined for the position of mate.[14] Since this would only apply to newly appointed officers it would depend on retirements for all the maritime service to be so qualified.

Conclusion

The specific mention by Captain Charles Mears in his log on board HEICS *Egmont* while at Tenerife of a timekeeper in March 1770 is probably the earliest record of the use of a timekeeper on a regular commercial voyage following the trial of John Harrison's H4. Furthermore, by 1773 there were at least two East India Company captains using a timekeeper to determine longitude. The circumstantial evidence indicates it was a John Arnold timekeeper that Charles Mears had on board in 1770. It clearly illustrates that certain East India Captains were taking serious interest in this new technology well in advance of dates previously identified and well ahead of the Royal Navy.

Mears's interest in timekeepers seems to have arisen as he was one of two captains to first use the new technology of the lunar-distance method for determining longitude on a long voyage in 1763. The influence of captains Mears and Dewar clearly created a growing interest in this method by other East India Company captains and ten years later 50 per cent of East India Company ships were using it on their two-year voyages.

14 Sutton, East India Company's Maritime Service, 132, note 21; British Library, India Office Records, L/MAR/C/1

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http://dx.doi.org/10.1080/00253359.2016.1210326
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References

Arnold, J., *Certificates and Circumstances Relative to the going of Mr Arnold's Chronometers* (London, 1791)
Betts J., 'Arnold and Earnshaw: The Practicable Solution', in W. J. H. Andrews (ed.), *Quest for Longitude* (Cambridge, Mass, 2013)
Gould, R. T., *The Marine Chronometer : Its history and development: incorporating Gould's own amendments and additions from his original annotated manuscripts* 2nd edn with new illustrations and foreword by J. Betts (Woodbridge, 2013)
Howse, D., 'The Lunar Observation Method', in W. J. H. Andrews (ed.), *The Quest for Longitude: The Proceedings of the Longitude Symposium, 1993* (Cambridge MA, 1996)
Maskelyne, N., *The British Mariners' Guide containing Complete and Easy Instructions for the Discovery of the Longitude at Sea and Land* (London, 1763)
Mercer, V., *John Arnold & Son Chronometer Makers 1763–1843* (London, 1972)
Sutton J., *The East India Company's Maritime Service 1746–1834*, (Woodbridge 2010)
Waddington, R., *A Practical Method for finding the Longitude and Latitude of a Ship at Sea by Observations of the Moon* (London, 1763)

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